

First Demonstration of a Zero Dispersion Hollow-Core Fibre with Telecoms-Grade Loss (0.2 dB/km) in the C-Band

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Abstract We demonstrate the first optical fibre combining ultralow nonlinearity with zero chromatic dispersion in the C-band at telecom-grade loss (~ 0.2 dB/km). Enabled by membrane-thickness engineering in a hybrid-window antiresonant hollow-core structure, the fibre unlocks new opportunities for optical communications, for both IM-DD and coherent transmission. ©2026 The Author(s)

Introduction

Hollow-core fibres (HCFs) have emerged as one of the most promising transmission media for future optical communication systems, offering a unique combination of ultra-low attenuation, reduced latency and dramatically lower nonlinear impairments compared with conventional silica single-mode fibres (SMFs) [1,2]. Recent advances have also demonstrated that fibres with Double Nested (DNANF) design can attain attenuations below that of SMF [3,4,5]. The chromatic dispersion (D) of these low-loss HCFs is also intrinsically lower than the ~ 17 ps/nm/km of standard SMF. For HCFs with core diameters of 25, 30 and 35 μm , D is typically ~ 4.8 , 3.4 and 2.5 ps/nm/km at their centre-band low-loss region, respectively, owing to an inverse relationship between dispersion and core size [6]. This already enables improved performance for low cost and

low power consumption intensity-modulation direct-detection (IM-DD) links. For example, a large core (36 μm), low dispersion (2.1 ps/nm/km) NANF could transmit 100 Gb/s over 38 km using PAM4 modulation at the 20% FEC limit, five times longer reach than allowed by SMF [7]. The absence of nonlinearity in HCFs enables higher received OSNR than with glass-cored fibres, so that the ultimate IM-DD performance is limited by the power-fading induced by the chromatic dispersion. The realization of a practical zero-dispersion HCF, therefore, combining the latency, loss and nonlinear advantages of hollow-core guidance with a transmission medium free from chromatic-dispersion penalties would represent a major milestone. It would allow extended reach IM-DD transmissions and potentially simpler DSP and lower power consumption in coherent transmissions, with significant economic benefits.

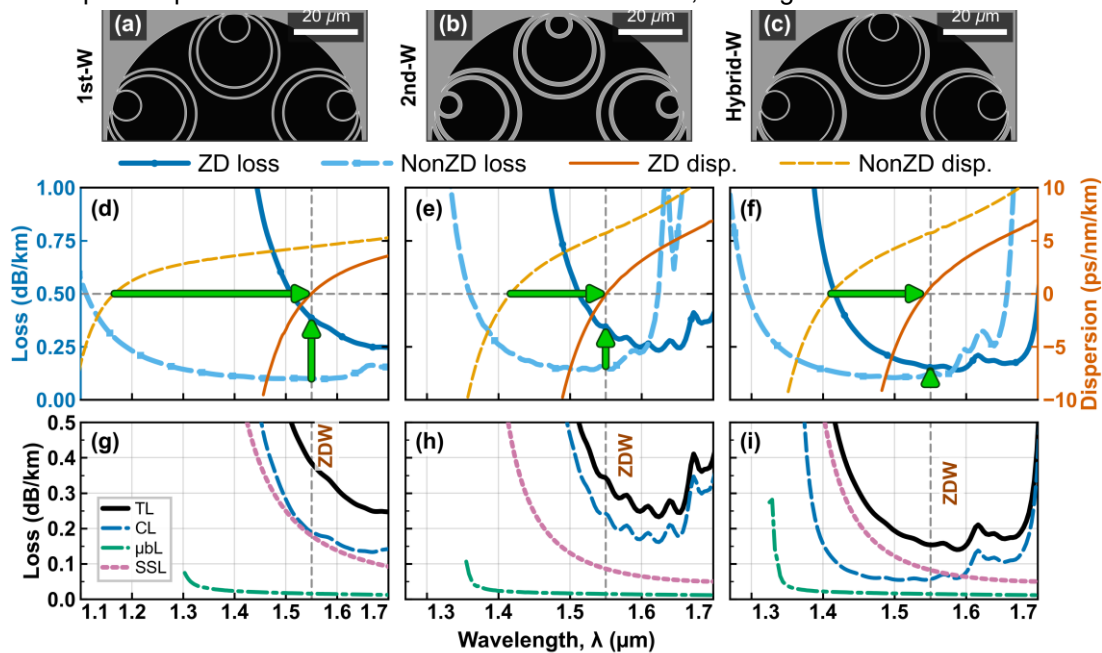


Fig. 1: First-, second-, and hybrid-window DNANFs (left, middle and right columns). All three fibres have a 26 μm core and identical tube diameters and positions; only their wall thickness changes. (a)–(c) Half cross-sections of the ZD geometries. (d)–(f) Total loss (blue) and dispersion (orange/yellow) for the NonZD and ZD variants; (g)–(i) loss contributions for the ZD designs: total (TL), confinement (CL), microbend (μBL), surface scattering (SSL).

Achieving near-zero dispersion in *low-loss* HCFs is challenging because their spectral dispersion profile follows a well-known cotangent shape, where the zero-dispersion wavelength (ZDW) occurs at wavelengths much shorter than mid-band [8,9], where loss tends to be high. In this work, we demonstrate that a **hybrid-window HCF design** [10] with precise tuning of the anti-resonant membrane thickness can enable zero-dispersion HCFs with a modest increase in loss versus the absolute lowest loss design. We explain the theoretical framework governing this behaviour, report fabrication of a set of seven fibres with systematically varied membrane thickness, and present two fibres exhibiting near-zero dispersion in the C-band while maintaining an attenuation of 0.2-0.23 dB/km. Finally, we experimentally validate the benefits of this approach through transmission over 15 km of fibre, demonstrating improved IM-DD performance relative to both standard SMF and conventional low-loss HCFs.

Design and Simulation

Fig. 1 compares simulations of three DNANFs, using loss models from Ref [11]. They all have the same cladding with five nested tubes, a 26 μm core diameter and a 3.5 μm inter-tube gap. Their different tube thickness make them operate in the *first* (Fig. 1a), *second* (1b) or *hybrid* (1c) window. For each design, we compared the fibre with the lowest-loss at 1550 nm (**NonZD**) with a version where the tube thickness was increased to shift the zero-dispersion to 1550 nm (**ZD**).

In the first-window DNANF, with tubes of ~ 450 nm thickness, the distance between the ZDW at 1166 nm and lowest loss wavelength (LLW) in the C-band is large (~ 400 nm). Bringing the ZDW to 1550 nm requires all three membranes to be up-scaled by $\sim 33\%$, red-shifting the entire loss curve. As a result, the total loss at 1550 nm increases from 0.10 to 0.39 dB/km (Fig. 1d), a

rather high and impractical loss penalty.

In the second-window DNANF, the antiresonance window is narrower, bringing the ZDW and the LLW closer. Thickening the membranes from 1180 nm to an average of 1293 nm does place the ZDW near 1550 nm (Fig. 1e). Since this brings operation close to the high-loss band edge, the penalty is still severe, rising from 0.16 to 0.34 dB/km at 1550 nm. This increase is predominantly due to confinement loss, accounting for 71 % of the total loss (Fig. 1h), showing that the fibre struggles to keep the mode confined at the ZDW.

In the hybrid DNANF, the outer tubes operate in the second window while the nested tubes remain in the first. Here, the low loss design has outer and nested tube walls of ~ 1180 and ~ 400 nm, respectively. The ZD design is realized through a modest ($\sim 10\%$) increase in only the outer thickness. The ZDW moves from 1411 nm to 1550 nm, yet the loss at 1550 nm only increases from 0.13 to 0.16 dB/km, a much smaller penalty than in the other two designs (Fig. 1f). Fig. 1i explains this behaviour: because the nested and outer resonators are antiresonant at different wavelengths, the hybrid structure forms a broad, shallow confinement-loss well. Shifting it barely raises the loss at the new ZDW.

Fabrication and Experimental Validation

To validate the modelling predictions, we fabricated six new hybrid DNANFs in addition to the reference low-loss fibre (HF-ref) reported in [10]. The fibres were produced from very similar mother preforms. We targeted geometrically similar structures with inter-tube gaps of 3.1 μm , and middle/inner tubes of ~ 500 nm wall thickness and ~ 25 $\mu\text{m}/\sim 10$ μm diameters. Their lengths ranged from 10.0 to 37.1 km.

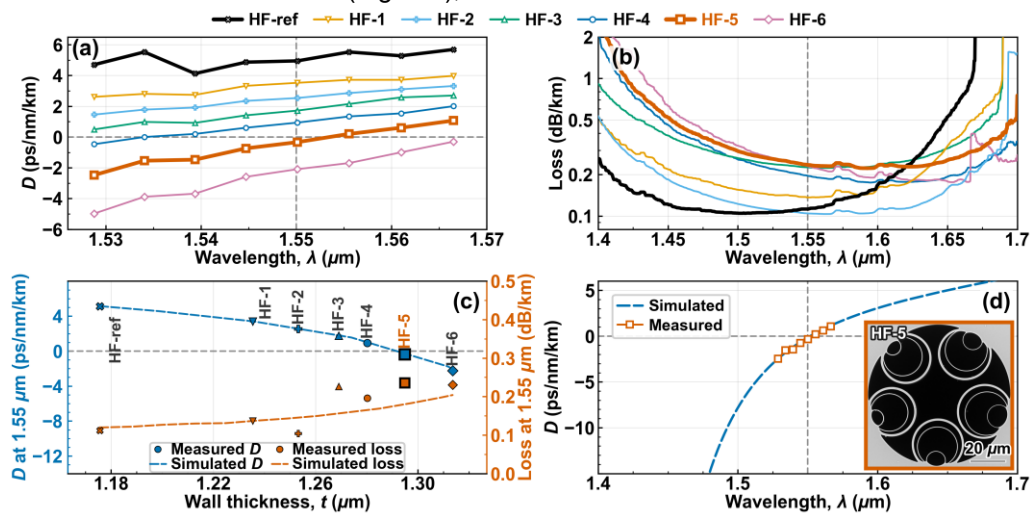


Fig. 2: Seven fabricated hybrid DNANFs: (a) Measured chromatic dispersion (D); (b) Cutback loss; (c) D and loss at 1550 nm versus outer-wall thickness t , compared with simulations. (d) Measured HF-5 dispersion compared with a polarisation-averaged simulation from the SEM; inset, SEM of HF-5 cross-section.

For all fibres we chose a small core diameter ($\sim 26 \mu\text{m}$) enabling compact fibres with relatively small ($300 \mu\text{m}$) coated diameters. To achieve dispersions spanning normal to anomalous regimes, we increased the thickness of the outer tube from ~ 1180 to 1320 nm in small, controllable steps.

Fig. 2a shows the C-band dispersion of the 7 fibres, measured by a 400G ZR pluggable hosted by an optical network tester. At 1550 nm we obtained a monotonic decrease in D , from $+5.1 \text{ ps/nm/km}$ of the 0.11 dB/km HF-ref to -2.2 ps/nm/km of HF-6. Simultaneously, the dispersion slope increased from 0.02 to $0.11 \text{ ps/nm}^2/\text{km}$. HF-5 has a ZDW at 1554.0 nm , with D increasing linearly from -2.3 ps/nm/km at 1530 nm to 0.6 ps/nm/km at 1560 nm . HF-4, fabricated from the same preform but with a 15 nm thinner outer tube, has a ZDW at 1535.4 nm .

Thickening the tubes red-shifts all the loss curves (Fig. 2b) with only a modest increase, from 0.11 to 0.23 dB/km at 1550 nm . At its ZDW, HF-5 has a loss of 0.23 dB/km ; HF-4 of 0.196 dB/km . HF-2 has an all-anomalous dispersion, from 1.5 to 3 ps/nm/km in the C-band, with a loss of only 0.11 dB/km , while HF-6 has an all-normal dispersion.

Fig. 2c shows the measured dispersion and fibre loss at 1550 nm as a function of the outer-tube wall thickness. The measured chromatic dispersion is in very good agreement with the simulations, with zero dispersion occurring at a wall thickness of approximately 1293 nm . At 1550 nm , a 10 nm change in outer-capillary wall thickness increases the dispersion by approximately 0.51 ps/nm/km and shifts the ZDW by 12.5 nm .

Simulations for ideal fibres indicate an increase of approximately 0.1 dB/km from the low-loss fibre design (HF-ref) to the zero-dispersion

fibres (HF-4,5). The measured loss shows some deviations from the simulated trend and a larger loss increase, which may be attributed to variations in fibre core diameter and/or outer-tube gap size, likely addressable with further engineering.

Fig. 2d shows the dispersion curve simulated from the SEM cross-sectional image of the fabricated HF-5 fibre (Fig. 2e), demonstrating excellent agreement with the measured chromatic-dispersion spectrum across the C-band.

IM-DD Data Transmission Comparison

Next, to evaluate the improvement these fibres can bring to dispersion-induced power fading, we performed an IM-DD transmission experiment comparing HF-5 with a standard HCF (similar core size) and an SMF-28. All were 15 km long. We used root-raised cosine filter-shaped 112-Gb/s 4-ary pulse amplitude modulation (PAM4, 56 GBaud , roll-off factor of 0.1). The bandwidths of transmitter and receiver were both well above the signal's bandwidth. The same decision feedback equalizer with 70 feedforward and 15 feedback taps was adopted for all cases. The BER results at around 1528.77 , 1550.12 , and 1566.51 nm (i.e., two edges and one middle C-band wavelengths) are shown in Fig.3, together with the eye diagrams at 1550.12 nm . Despite the relatively short transmission length ($\sim 15 \text{ km}$), the signal cannot be recovered in the SMF case due to two severe frequency nulls falling inside the signal's bandwidth. As expected, the conventional HCF offers much better performance relative to SMF. The near-zero dispersion of HF-5 further reduces the BERs by around one order of magnitude, highlighting its potential for high baud rate/longer distance IM-DD transmission.

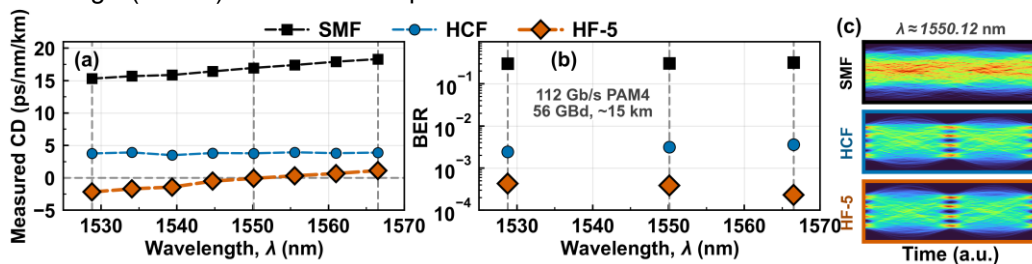


Fig. 3: (a) measured dispersion of the 3 fibres under test. (b) BER of 112 Gb/s PAM4 transmission through 15 km of the three fibres; (c) corresponding eye diagrams at 1550.12 nm .

Conclusions

We have shown for the first time that effective chromatic dispersion tuning is possible in an HCF by controlling its outer tube thickness. In DNANFs with hybrid-window guidance, such dispersion tuning only adds a modest loss penalty versus the lowest loss designs. To illustrate the potential, we produced two fibres with zero dispersion in the C-band: at 1535 (HF-4) and 1554 nm (HF-5), with loss of 0.2 and 0.23 dB/km , respectively. We also fabricated a fibre with only 0.11 dB/km and

reduced dispersion behaviour (HF-2); and another with an all-normal dispersion in the C-band (HF-6). All the HCFs produced here have a small core diameter of $26 \mu\text{m}$, but our simulations indicate that zero-dispersion at 0.10 dB/km should be possible with slightly larger ($29 \mu\text{m}$) cores. After ultra-low nonlinearity, latency and loss, carefully engineered HCFs can now achieve an ultra-low (or zero) chromatic dispersion, unlocking new possibilities to reduce power consumption and improve performance in many applications.

Acknowledgements

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